Lecture 2

Demand and Supply

ECO 101: Introduction to Microeconomics

Fall 2026

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Roadmap

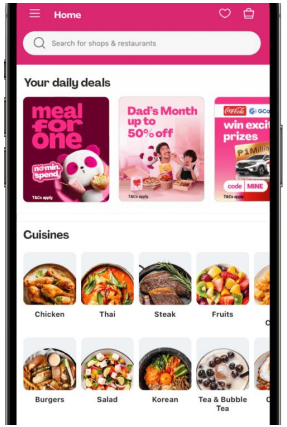
Lectures:

1. Production Possibility Frontier & Gains from Trade
2. **Law of Demand and Supply**
3. Elasticity
4. Government Actions in Markets
5. The Production Process: inputs & outputs
6. Perfect Competition
7. Monopoly
8. Game Theory

What is a Market?

A market is an arrangement through which **buyers** and **sellers** interact to exchange goods and services.







- The buyers as a group determine the demand for the product.
- The sellers as a group determine the supply of the product.

What is Demand?

Demand is the willingness and ability of consumers to buy a good or service at different prices.

To have effective demand for a good or service, three conditions must be met:

- Want it: You desire the good or service.
- Able to afford it: You can buy it.
- Willing to buy it: You intend to purchase it at a given price.

Kacchi & Transport on Rainy Days





6 main factors that affect Demand

1. **Income**: The level of income affects our demand for certain goods or services.

- Normal good: an increase in income leads to an increase in demand
- Inferior good: an increase in income leads to a decrease in demand

2. **Price of Related Goods**: A change in the price of one good affects the demand of a good that may serve as a complement or a substitute to it.

6 Main Factors that Affect Demand (contd.)

3. **Population**: When population increases, so does demand for goods and services.

4. **Tastes & Preferences**: Tastes and preferences can change over time and lead to our demanding more of a good or service while preferring less of another.

5. **Seasonality**: Certain goods and services have seasonal demands. Their demands fluctuate based on some easily predictable and intuitive factors.

6. **Expected Future Prices**: If there's a general expectation of the price of a good to increase in the future, consumers are more likely to stock up on that particular good.

Demand vs. Quantity Demanded

Demand and Quantity Demanded are not the same.

Demand is impacted by the 6 main factors we just discussed. These factors change how much of a product is demanded at every single price point.

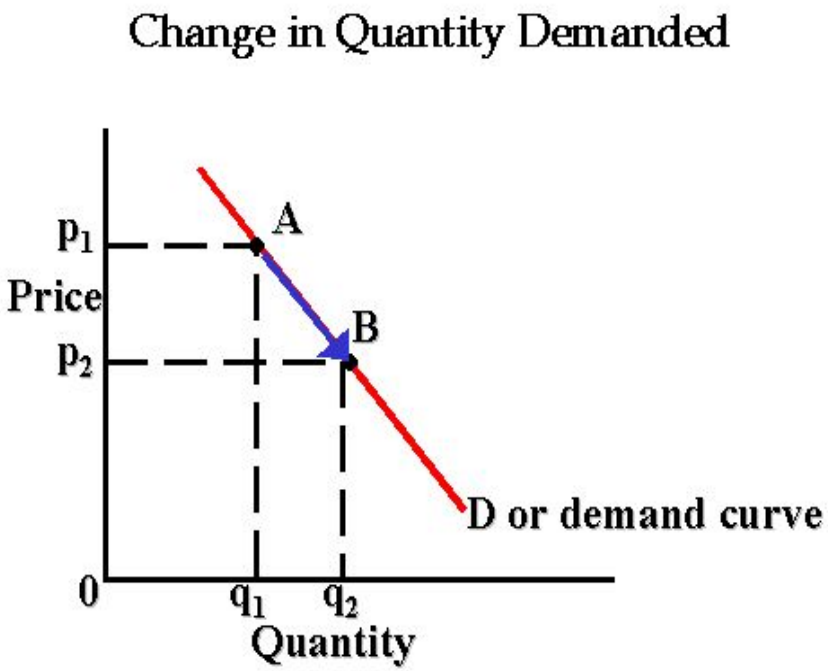
Quantity demanded is the amount of a good that buyers are willing and able to purchase. The only factor that directly changes the quantity demanded of a product is its **own price.**

Law of Demand

The price of a good or service is inversely related to a change in the quantity demanded of that good or service.

A rise in the price of a good or service will, *ceteris paribus*, lead to a fall in the quantity demanded of that good or service, and vice versa.

Change in Quantity Demanded

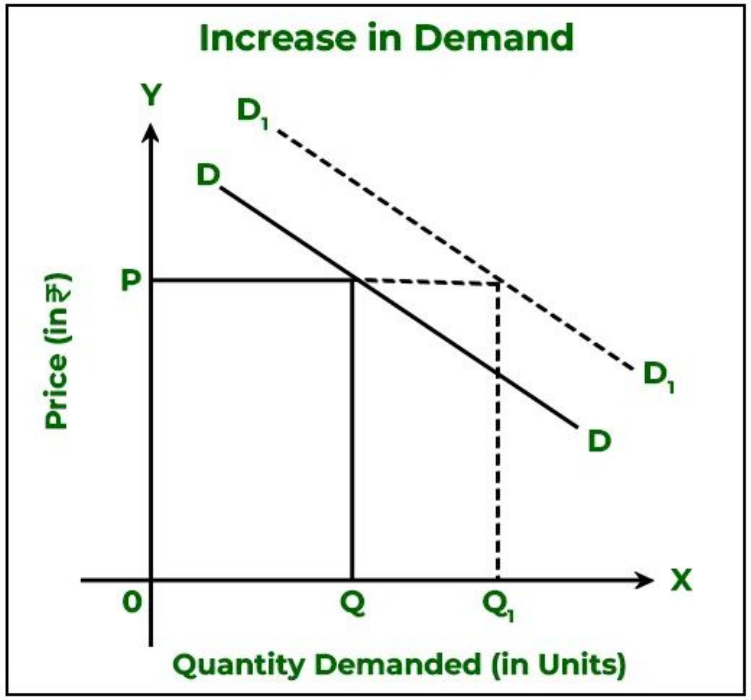


- ➔ Movement along the curve
- ➔ Result of a change in the price
- ➔ Other factors remain constant

The price falls from p_1 to p_2, which leads to an increase in quantity demanded (q_1 to q_2) along the demand curve.

Change in quantity demanded lead to a movement along the point from A to B.

Change in Demand



- ➔ A shift in the demand curve
- ➔ Price remains constant
- ➔ One of the six factors change

The initial demand curve is D. It shifts to the right. The new demand curve is now D_1.

The new quantity demanded is Q_1.

Quick Quiz on Demand

1. The price of Matcha falls from Tk. 500 to Tk. 450. What happens?

- Demand shifts right
- Quantity demanded increases
- Demand shifts left

2. A celebrity wears the new Nike sneaker and it goes viral. What happens?

- Movement along demand
- Demand shifts left
- Demand shifts right

Demand Curve

The Law of Demand states that as the price of a good or service falls, the quantity demanded of that good or service increases. When the demand curve is at a straight line, the following equation describes it:

$$P = a - bQ_D$$

Here, P is the price and Q_D is the quantity demanded. a and b are positive constants.

What does the Demand Equation tell us?

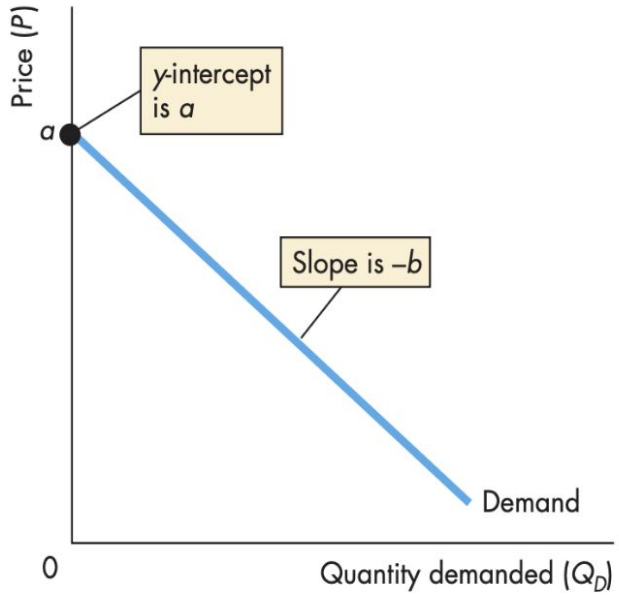


Figure 1 Demand curve

- The price at which no one is willing to buy the good is when the price is at a. The quantity demanded at this price is zero. The price at which the demand curve hits the y-axis can be regarded as the demand curve's y-intercept.

- The quantity demanded increases as the price falls. If Q_D is a positive number (> 0), then the price, P, must be less than a. The larger the Q_D grows, the smaller the price becomes, i.e., *if the quantity increases, the maximum price that buyers are willing to pay for the last unit of the good falls.*

- The constant b tells us *how fast the maximum price that someone is willing to pay for the good falls as the quantity increases*. The constant b tells us the *steepness of the demand curve*. The equation tells us that the slope of the demand curve is –b.

What is Supply?

There's a positive relationship between price and quantity for suppliers, as opposed to the inverse relationship for consumers, i.e. consumers want to buy a good or a service at the cheapest price possible, while producers want to sell a good for the highest price possible.

The 3 requirements to be fulfilled for a good or service to be supplied to the market:

- Access to the necessary resources & technology to make the good
- Able to generate profit in the market
- Plan to produce and supply

4 main factors that affect Supply

1. **Cost of the factors of production**: The producer introduces inputs—labour, raw materials, machinery, capital, etc.—and gets an output in the process. These inputs are collectively called factors of production. These factors have associated costs—wages, rent, interest payments if the machinery is on loan, among others. If the cost of one of them goes up, the cost of the production goes up. This can adversely impact the supply, and vice versa.

4 main factors that affect Supply (contd.)

2. **Price of related goods in production**: During the production process, there are certain goods that are substitutes in production (using almost the same raw materials) and complements in production (arise out of the same production process). A rise in the price of one good will lead to a fall in the supply of the substitute (as more of the raw materials will be used to produce the good that is priced higher in the market) even though the price remains the same. On the other hand, for complementary goods, a rise in the price of one good will lead to a rise of the supply of the other good.

Examples: Carbonated drinks & energy drinks, beef & leather, butter & skimmed milk, plastic toys & plastic household containers.

4 main factors that affect Supply (contd.)

3. **Number of Suppliers**: More suppliers/producers in the market means more supply of goods and resources. However, goods will only be supplied to the market if the producer expects to generate profit. Once all demand is met, there is no motivation for additional suppliers. So, there is an upper bound on the number of suppliers in any given market.

4. **Technology**: When there are technological advances, we can produce the same goods at a lower cost or more with the same amount of resources. This leads to an outward shift of the PPF which indicates a rise in the supply.

Supply vs. Quantity Supplied

Quantity Supplied is the amount of a good that sellers are willing and able to sell.

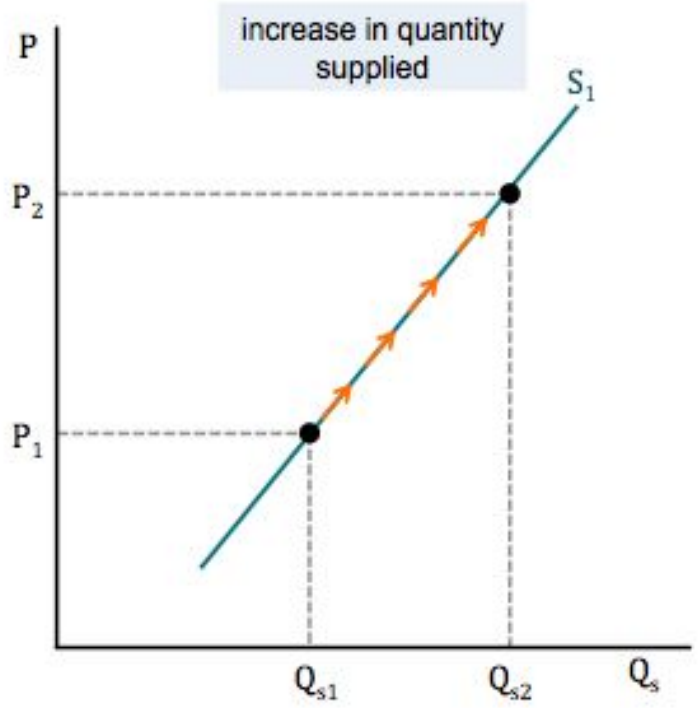
If we look at the other factors that have been discussed, the price level of the goods remains the same. At the same price level, because of a change in one of the factors (e.g., rise in labour wages), the quantity of goods supplied goes down. This is *a fall in supply, not a fall in quantity supplied*.

Law of Supply

The price of a good or service is positively related to a change in the quantity supplied of that good or service.

When the price of a good changes, suppliers adjust the quantity they are willing to sell to maximise profit. However, other factors can also affect supply.

Change in Quantity Supplied

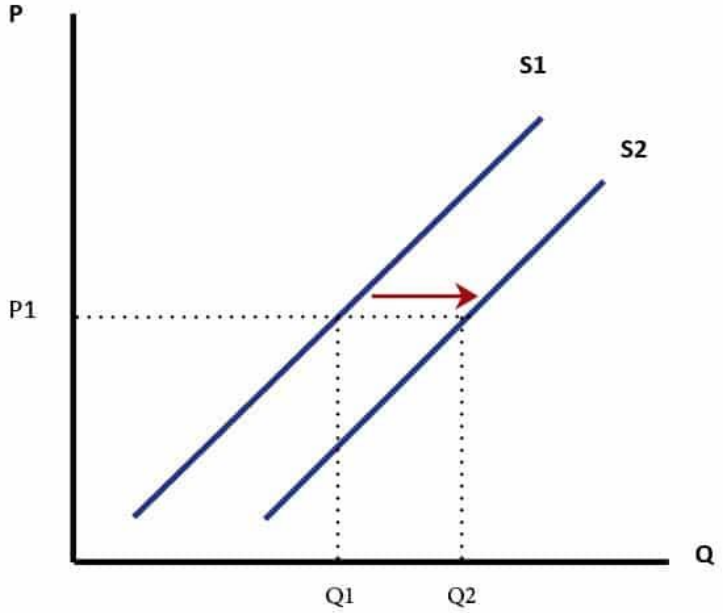


- ➔ Movement along the curve
- ➔ Result of a change in the price
- ➔ Other factors remain constant

The price rises from P_1 to P_2, which leads to an increase in quantity supplied from Q_{s1} to Q_{s2}.

Change in quantity supplied lead to a movement along the supply curve.

Change in Supply



- ➔ A shift in the supply curve
- ➔ Price remains constant
- ➔ One of the four factors change

The initial supply curve is S_1 which shifts to the right. The new supply curve is now S_2.

The new quantity supplied is Q_2.

Quick Quiz on Supply

1. A bakery's cost of flour increases. What is the most likely effect on the supply of bread, other things remaining equal?

- Supply curve shifts to the right
- Quantity Supplied rises along the supply curve
- Supply curve shifts to the left

2. A café increases the price of its iced coffee from Tk.120 to Tk.180 per cup. As a result, the café decides to offer more cups for sale each day. Assuming other factors remain constant, which of the following best describes this change?

- Quantity supplied decreases
- Quantity supplied increases
- Supply curve shifts to the right

Supply Curve

The Law of Supply states that as the price of a good or service rises, the quantity supplied of that good or service increases. When the supply curve is a straight, positively shaped line, the following equation can describe it:

$$\boxed{P = \boldsymbol{c} + dQ_S}$$

Here, P is the price and Q_S is the quantity supplied, and c and d are positive constants.

What does the Supply Equation tell us?

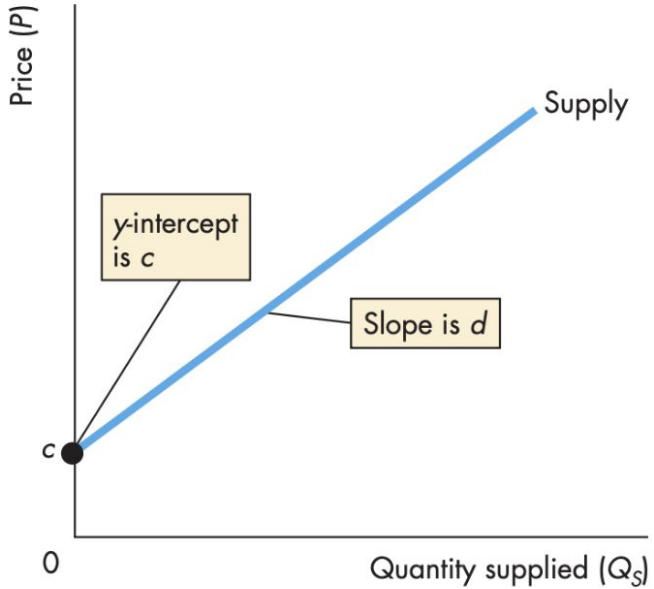


Figure 2 Supply curve

- The price at which sellers are not willing to supply the good is the price c in this figure. The price c is where the supply curve hits the y-axis. It can be regarded as the supply curve's y-intercept.

- As the price rises, the quantity supplied increases. If Q_S is a positive number, then the price P must be greater than c. As the quantity increases, the minimum price that sellers are willing to accept for the last unit rises.

- The constant d is the slope of the supply curve, as per the equation. It tells us how fast the minimum price at which someone is willing to sell the good rises as the quantity increases. That is, the constant d tells us about the steepness of the supply curve.

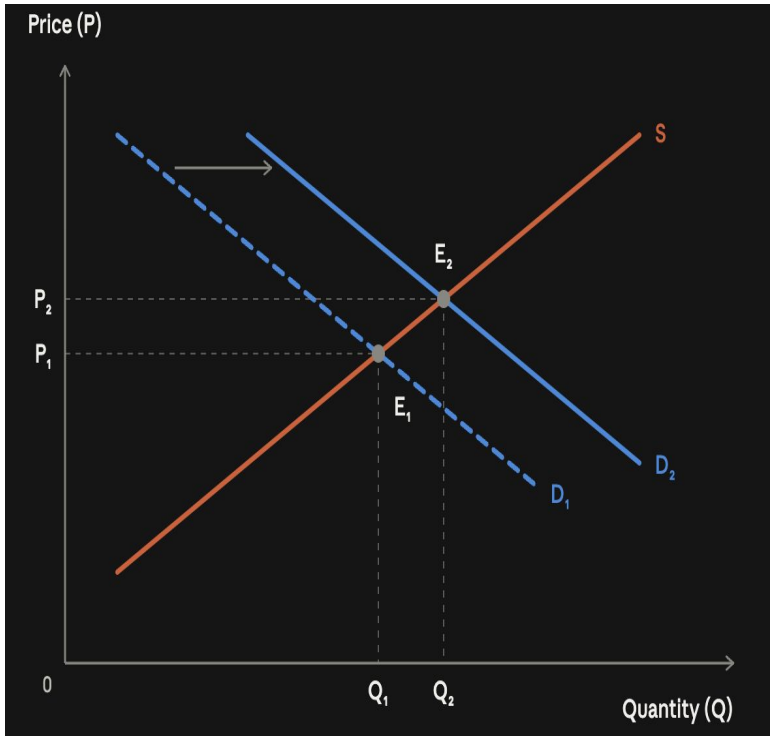
Market Equilibrium

A situation in which the market price has reached the level at which quantity supplied equals quantity demanded is called market equilibrium.

Equilibrium is a point at which there is no further motivation for anyone involved to change anything. So far, there are two economic agents that we have discussed: the consumers and the producers. At the equilibrium price, the quantity of a good that buyers are willing and able to buy exactly balances the quantity that sellers are willing and able to sell.

The market equilibrium will be at a point at which, given the price level, the quantity demanded is equal to the quantity supplied.

Market Equilibrium: graph (shift in demand curve)

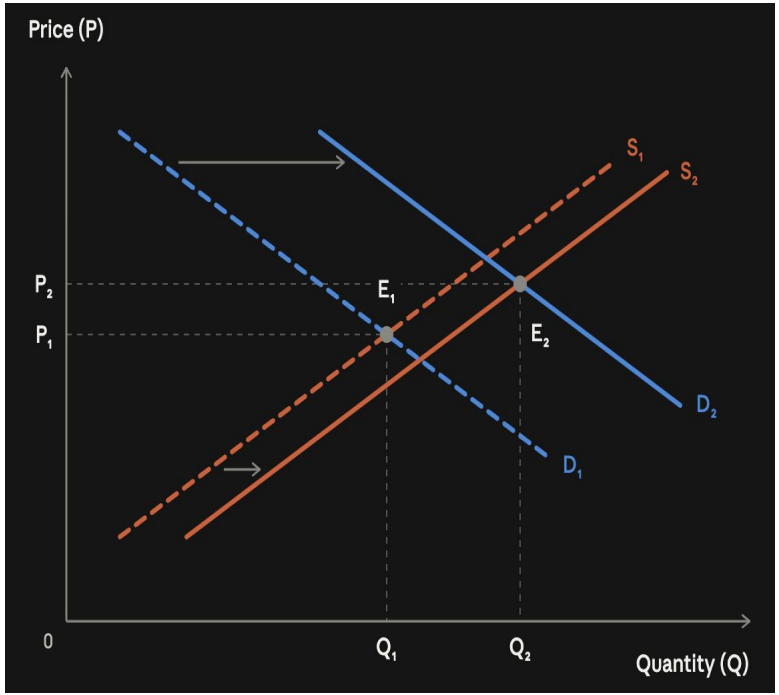


Here, the initial demand curve D_1 and the supply curve S, intersect at a point E_1, which is the equilibrium.

A shift in either of these curves will lead to a new equilibrium.

- ➔ An increase in demand (e.g. due to a rise in income) leads to a right shift of the demand curve from D_1 to D_2.
- ➔ This gives rise to the new equilibrium, E_2.
- ➔ At this point, both price (P_1 to P_2) and quantity (Q_1 to Q_2) are higher.

Market Equilibrium: graph (shift in demand & supply curves)



Here, the initial demand curve D_1 and initial supply curve, S_1, intersect at a point E_1, which is the equilibrium.

- ➔ Demand and supply both increase. The demand curve shifts right from D_1 to D_2. The supply curve shifts right from S_1 to S_2.
- ➔ The new equilibrium is at E_2.
- ➔ **Quantity:** It rises from Q_1 to Q_2 whenever demand and supply both increase.
- ➔ **Price:** In this graph, it rises, because the demand shift is larger than the supply shift. If the supply shift were larger, price would fall. If the two shifts were equal, price would stay the same.

Market Equilibrium (contd.)

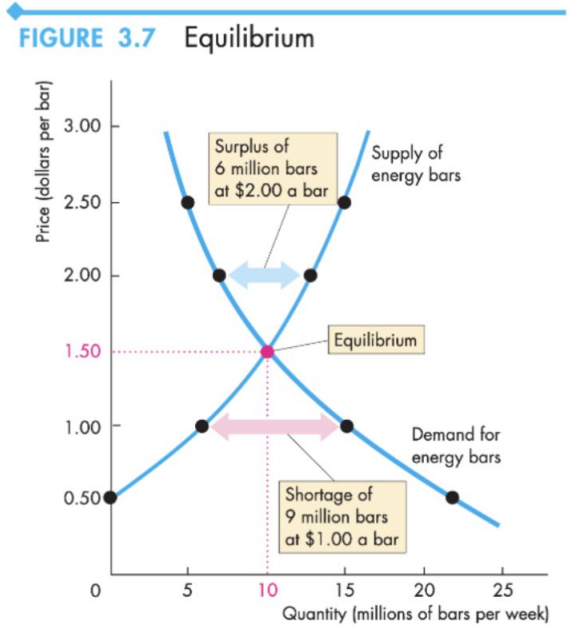
An equilibrium is a situation in which opposing forces balance each other. Equilibrium in a market occurs when the price balances buying plans and selling plans. The equilibrium price is the price at which the quantity demanded equals the quantity supplied. The equilibrium quantity is the quantity bought and sold at the equilibrium price.

A market moves its equilibrium because:

- Price regulates buying and selling plans.
- Price adjusts when plans don’t match.

Price as a regulator

The price of a good regulates the quantities demanded and supplied. If the price is too high, the quantity supplied exceeds the quantity demanded. If the price is too low, the quantity demanded exceeds the quantity supplied.



Price (dollars per bar)	Quantity demanded (millions of bars per week)		Quantity supplied (millions of bars per week)
	Original	New	
0.50	22	32	0
1.00	15	25	6
1.50	**10**	20	**10**
2.00	7	17	13
2.50	5	**15**	**15**

Mathematical Note

To find the equilibrium price and equilibrium quantity, substitute Q^* for Q_D in the demand equation and Q^* for Q_S in the supply equation. Then the price is the equilibrium price (P^*), which gives

$$P^* = a - bQ^*$$
$$P^* = c + dQ^*.$$

Notice that

$$a - bQ^* = c + dQ^*.$$

Now solve for Q^*:

$$a - c = bQ^* + dQ^*$$
$$a - c = (b + d)Q^*$$
$$Q^* = \frac{a - c}{b + d}.$$

To find the equilibrium price, (P^*), substitute for Q^* in either the demand equation or the supply equation.

Using the demand equation, we have

$$P^* = a - b\left(\frac{a - c}{b + d}\right)$$
$$P^* = \frac{a(b + d) - b(a - c)}{b + d}$$
$$P^* = \frac{ad + bc}{b + d}.$$

Alternatively, using the supply equation, we have

$$P^* = c + d\left(\frac{a - c}{b + d}\right)$$
$$P^* = \frac{c(b + d) + d(a - c)}{b + d}$$
$$P^* = \frac{ad + bc}{b + d}.$$

Calculation of Equilibrium Price & Quantity

The demand for ice-cream cones is

$$P = 800 - 2Q_D.$$

The supply of ice-cream cones is

$$P = 200 + 1Q_S.$$

The price of a cone is expressed in cents, and the quantities are expressed in cones per day.

To find the equilibrium price (P^*) and equilibrium quantity (Q^*), substitute Q^* for Q_D and Q_S and P^* for P. That is,

Write down the Demand equation and the Supply equation considering that the demand and supply curves intersect at the equilibrium. Find Q^* and P^*.

$$P^* = 800 - 2Q^*$$

$$P^* = 200 + 1Q^*.$$

Now solve for Q^*:

$$800 - 2Q^* = 200 + 1Q^*$$

$$600 = 3Q^*$$

$$Q^* = 200.$$

And

$$P^* = 800 - 2(200)$$

$$= 400.$$

The equilibrium price is \$4 a cone, and the equilibrium quantity is 200 cones per day.

Exit ticket

Before you leave the class today, answer the following:

1. What happens to quantity demanded when the price of a good increases, other things remaining equal?
2. How does an increase in consumers' income affect the demand for normal goods and inferior goods?
3. If quantity demanded exceeds quantity supplied at the current market price, what happens in the market?
4. What is market equilibrium?
5. How does price act as a regulator?